Satchel Schiavo

sschiavo@olin.edu https://sites.google.com/view/satchelschiavo/home

Education

Olin College of Engineering | Bachelor of Science in Mechanical Engineering | GPA 3.6 | 2022 - 2026, NEEDHAM, MA

Awards

Recipient of 50% Merit Scholarship | Class Representative (2023 - Current) | R2 (RA Equivalent)

Skills

Software: SolidWorks, OnShape, ANSYS Fluent, COMSOL, MATLAB, Python, ROS2, Java, R, HTML, Arduino IDE

Manufacturing: CNC Mill/ Lathe, Water Jet, Laser Cutting, 3D Printing, Sheet Metal, Manual Mill/Lathe, Injection Molding, TIG Welding

Experience

Amazon Robotics / ROBOTICS ENGINEER - SENIOR CAPSTONE — SEPTEMBER 2025 - MAY 2026, NORTH READING, MA

- Selected as one of five senior engineering students to work with Amazon Robotics R&D in year-long hybrid project
- Evaluated six perception architectures using LiDAR, stereo vision, RGB-D cameras, and monocular depth estimation Designed testing framework to benchmark depth accuracy, object tracking, environmental robustness, and system cost
- Developed ROS2-based perception pipeline integrating multi-sensor data streams for real-time autonomous navigation

Berkshire Grey / HARDWARE ENGINEERING INTERN — JUNE - AUGUST 2025, BEDFORD, MA

- Leveraged mechanical design and rapid prototyping of a robotic package-alignment test platform, collaborating with a 12-engineer cross-disciplinary team from concept through validation.
- Redesigned conveyor subsystem using Design for Manufacturing principles, reducing package spillage to below 0.1%
- Authored production documentation including BOMs, assembly drawings, and maintenance procedures

Olin Baja SAE / MECHANICAL DESIGN LEAD — SEPTEMBER 2022 - PRESENT, NEEDHAM, MA

- Developed MATLAB suspension kinematics model to evaluate camber, caster, toe, scrub radius, kingpin inclination, and Ackermann characteristics, designed a moving suspension study for the front A-arms to simulate movement
- Managed Full Car CAD for 24-25 competition cycle, reorganized part structure to improve coordination and design goals
- Optimized geometry (camber, caster, toe, SAI, scrub radius) improving contact patch engagement, ease of handling; welded rear diff to increase traction; repositioned trailing arms to achieve Anti-Ackerman steering
- Led a subteam of 15 people using project management skills, mentored first-year students in redesigning cockpit to include top-mounted brake and gas pedals, taught Solidworks CAD and FEA to validate new design projects

Olin Electric Motorsports / AERODYNAMICS RESEARCH LEAD — SEPTEMBER 2024 - 2025, NEEDHAM, MA

- Co-founded the Aerodynamics subteam for Olin Formula SAE to design and test the aero package for their MK6 car.
- Designed front wing system and rear diffuser to improve downforce and handling at higher speeds using Solidworks and analyzed with SOLIDWORKS CFD and ANSYS.
- Manufactured first iteration of design, fabricated using hemp composites, heat wire, and insulation foam

KEEN Foundation / DATA SCIENCE INTERN — MAY 2023 - AUGUST 2024, REMOTE

- Analyzed two years' data from four partner universities in the KEEN college network on what aspects of intercollegiate research website were most impactful using Qualtrics, Excel, Python and RStudio, implemented changes in HTML
- Increased first-year awareness of undergraduate research by 31.5% across four universities, contributed to a 16% increase in research applications

Relevant Projects

Student Instructor - Mechanical Analysis, Spring 2026

Selected by Olin College's Academic Review Board to design and teach an accredited Mechanical Analysis course

Tetris Playing Robot - Computational Introduction to Robotics, Spring 2026

Owned CV integration on a robotic system that could play Tetris on a Nintendo Switch by reading the TV screen, analyze game state, and use a robot arm to manipulate a controller. Developed computer vision pipeline to identify game state from live video and generate autonomous control inputs through a robotic manipulator.